



UMEÅ UNIVERSITY

Designing for UX Designers

Creating Sustainable and Usable Design Systems

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Abstract

A design system is a framework that guides design and development teams through shared standards and practices. It consists of tangible and intangible elements. The tangible elements as guidelines, color palettes, components, and patterns, and intangible elements that reflect the organization's values and design principles. Together, these ensure consistency and efficiency across digital products. Without a design system, inconsistency arise, making it challenging to maintain a consistent user experience. Having a functional design system is crucial for UX designers when creating structured and consistent design solutions. While previous research about existing design systems often has a focus on developers' needs and technical implementation, there is limited research that address the practical needs and workflows of UX designer. This study investigates how a design system can be structured to support UX designers' everyday practices while ensuring long-term usability and maintainability. To answer this, a combination of Research through Design and Design Thinking methodology along with qualitative data collected from interviews and user tests with experienced UX designers. The findings indicate the importance of documentation, reusable components and the central role design systems have in a design process. These insights contribute to a practical understanding of how design systems can support sustainable UX work overtime.

Keywords: Design System, UX Design, Research Through Design, Design Thinking, Long-term Usability, Reusable Components, Documentation

1. Introduction

A design system is a collection of reusable components, guidelines and standards that guide the design and development of digital products. By having consistency and structure, design systems allow teams to remain efficient and collaborate effectively. In UX design, such frameworks support coherent rules for layout, color, typography and user flow, helping UX designers to create interfaces that are both functional and usable (Moore et al., 2020).

The concept of design systems builds on the idea of design patterns, reusable solutions to common design challenges. Christopher Alexander's theory of living structures and design patterns, encourage designers to create interfaces that evolve over time and that are responding to user needs (Gorichanaz, 2023). Design patterns improve the design process in the aspects of both time and effort (Mirning et al. 2016), however, while design is a process it is also a product, shaped by user needs and resources. By following design guidelines, a designer can understand user goals with principles grounded in how people perceive, remember and act, which in turn helps create interfaces that are not only functional but usable while supporting users efficiently (Johnson, 2020).

In addition to usability and efficiency, design systems must also support principles of sustainability. In this study, "sustainability" refers not to environmental concerns, but rather to long-term maintainability and adaptability. The ability of a design system to remain useful,

relevant, and efficient over time despite evolving user needs and technological changes. By incorporating these principles, designers can prevent mistakes and develop solutions that is effective and relevant in the future (Knowles et al. 2016).

Design systems are often presented as bridges between design and development in existing research. They are recognized for their ability to improve reusability, simplify content creation, and strengthen collaboration between UX designers and developers (Moore et al., 2020). However, most research focuses on the developer's perspective, leaving the role and needs of UX designers underexplored. For example, a framework presented by Nokleberg & Hawkes (2021), emphasizing on productivity through code reuse, and Palomino's et al. (2024) study on how addressing developers' needs through user research and iteration can foster better adoption and collaboration. These are just a few of many studies highlighting the technical and developer-oriented benefits of design systems.

That said, these works show that the technical benefits of design systems are well established, with code efficiency, modularization, and integration proven to improve both productivity and implementation. It has been explored in Lamine & Cheng (2020) study, the need for better tools that facilitate the creation and further development of design systems. However, despite this identified need, there is still a lack of concrete solutions for how these tools should support a UX designer and developer in practice. From the perspective of this study is a core issue, mainly for how a design system can support a UX designer in their everyday work. Another study shows that although collaboration tools like Figma have changed the way UX designers work together, new collaboration problems arise, mainly between designers and developers. Feng et al. (2023) propose using text annotations in the design, to better connect design and code, as well as new ways of working where decision-making processes, rules and guidelines, and division of responsibilities govern how a design system is managed, developed, and used in an organization.

Altogether, it reveals a significant imbalance in emphasis where the everyday metrics, needs, and challenges of UX designers are still relatively underexplored. This indicates a critical gap with the absence of UX designers' perspectives. Because if design systems are primarily structured around the needs of developers, what role do UX designers play? With too much focus on the needs of developers, the system risks becoming less effective for UX designers, which in turn can hinder usability, adaptability, and long-term sustainability of the system.

When we expressed interest in exploring design systems from a UX designer's perspective, the tech company we are collaborating with saw it as an opportunity to also investigate which design choices and components are reusable, sustainable, and important in an evolving design system. Somewhere along the way in our exploration, we realized together with the company that it is more important to focus on investigating the needs of UX designers and what they consider to be the most essential elements a design system should include.

By focusing on UX designers' perspective, our study contributes to balancing the existing research field and provides insights into how design systems can be structured to meet UX designers' needs more effectively. When we refer to a "structured design system" in this study, we mean a systematically organized and well-documented framework consisting of reusable components, clear guidelines, and design principles.

As part of this research, a design system was developed and tested in collaboration with experienced UX designers, to explore how theoretical principles can be applied in practice. To address this, our research is guided by the following question: *How can a design system be structured to support UX designers' practical workflows while ensuring long-term usability and maintainability?*

To answer this research question, we have combined two frameworks, Research through Design and Design Thinking, to investigate and develop a design system. This methodological approach allows for an iterative and exploratory design process where the creation of design serves as a source of knowledge while are grounded in deep understanding of user's needs. The data collection was conducted through semi-structured interviews and user tests which provided us with insights into how design system supports usability and sustainable design practices.

This study is delimited in some important ways, when developing the design system, we made the choice to solely focus on desktop components and exclude mobile design considerations due to time constraints. The creation of the design system is conducted entirely in Figma, which also limits the study to the perspectives of UX designers rather than examining the needs or workflows of developers. As a result, our findings and conclusions are primarily to UX designers working in design systems. By involving UX designers throughout the process, we aim to provide a foundation for a design system that is not only usable, but adaptable and sustainable for future needs.

1.1 Aim and Research Question

The aim of this study is to explore how design systems can be developed in ways that support UX designers' everyday practices and long-term needs. By collaborating with experienced UX designers, this research seeks to identify key principles for creating a design system that can address both current and future needs, principles that, while developed in collaboration with Company X, are intended to be applicable and adopted across multiple companies. The central research question guiding this study is: *How can a Design System be developed to better align with UX designers' practical workflows while ensuring long-term usability and maintainability?*

1.2 Case setting

Our thesis has been conducted in collaboration with a tech consulting company in Umeå, referred to as Company X due to confidentiality and a non-disclosure agreement. Our study is more general and broadly applicable, we avoid directly associate our findings to Company X, encouraging focus on our analysis rather than the organization itself.

At the time of our study, Company X lacked a complete, reusable design system. We were granted access to an early prototype built in Figma through our supervisor, which served as a foundation for our work. In line with Company X's request, we explored which design choices and components are reusable, sustainable and essential in an evolving design system, resulting in a system shaped by UX designers' practical needs.

2. Related Research

When exploring a research topic, understanding existing literature and reviewing past research allows us to identify patterns and gain a deeper insight into the subject. In the following section we present an introduction to *Design Systems*, followed by a brief description of *Collaboration & Communication in Design Systems*, *Design Patterns*, *Design Guidelines* and *Reusable and Sustainable Design*. This section also addresses *AI & Automation in UX Tools* and *Perspectives on Design Systems*, as they play a significant role in the future aspect of a Design System.

2.1 Design System

The terminology for design systems can have slightly different meanings depending on the parties involved. For instance, a brand-focused company might describe a design system with brand guidelines, whereas in product development, a design system is understood as a framework that guides teams through common standards and practices. A design system is essentially a set of both tangible and non-tangible objects that together form systematic language. These tangible objects are guidelines, color palettes, button styles, patterns and UX/UI components. While the non-tangible objects capture the brand's values, beliefs and their best practices (Edelberg & Kilrain, 2020).

In user experience and interaction design, an understanding of human interaction is more central than focusing solely on graphic design. Design systems define the fundamental elements and components of visual and interactive design – layout, form, color, palettes, buttons, typography and user flows, while also ensuring the brand's tone and personality is consistent across products. By doing so, they create a cohesive and structured user experience across different interfaces (Moore et al., 2020).

Having a design system has become essential when achieving consistency and efficiency in software development. They act as a source for standards, reusable design patterns and components that support workflow and ensure cohesive outcomes (Myllylä, Karvonen, & Koskinen, 2024). Having a design system can minimize effort by ensuring consistency and reducing variation over products and product suites. It offers a foundation for learning, enabling teams to reconstruct, reorganize, expand, and adapt over time. Additionally, design systems encourage collaboration, enhancing teamwork between designers and developers (Moore et al., 2020).

Furthermore, Moore et al. (2020) explains a design system through three essential components: philosophy, interaction pattern and a content format. A design philosophy defines the fundamental vision and objectives of a system. Interaction patterns provide reusable components, forming the foundation of the user experience within a design system. Allowing designers to avoid rethinking user interactions and enabling users to engage with the application without the need to learn new interaction methods. A content format simplifies the creation of new content for specific scenarios. It empowers content designers to generate material for the application without needing to engage with the systems' inner workings. Meaning, that the process is simplified and more efficient, saving time and reducing complexity, allowing content designers to concentrate on content development rather than programming intents and dialog elements (Alao et al., 2022).

2.2 Collaboration & Communication in Design Systems

Companies are encouraged to think of a design system as a foundation for their future builds. While approaches vary, to achieve the best and most effective results possible design systems depends on structured processes, collaboration and communication within teams. Having consistency with guidelines and visual language is important, but teams that work together, deliver a much better and more cohesive experience for the end user (Edelberg & Kilrain, 2020).

According to Feng et al. (2023), digital tools that enable rapid feedback and continuous collaboration with diverse teams across disciplines has transformed the collaborative methods in UX practices. Tools like Figma have emerged, making it easier for professionals, UX designers, who work on designing and improving user experiences, while opening collaboration to non-designers such as developers.

However, research has shown that collaboration tools can create unexpected problems withing and outside a design team. This can indicate a lack of control, division of responsibilities, or technical misunderstandings. Still, tools like Figma have made it possible for professionals, UX designers, to collaborate within teams but also created new collaboration opportunities for developers. But it is with the advent of tools like Figma, which allow real-time collaboration, that multiple people can work on the same file at the same time. This has enabled new UX approaches such as design systems, where reusable design components are collected (Feng et al., 2023).

Feng et al. (2023) also suggest that there should be a fourth element beyond the three (components, patterns and guidelines) basic parts of a design system: governance handbook. This handbook should clarify responsibilities, describe how components can be changed or added and contain rules for how UX designers act as moderators for parts of the system. The handbook should have a structure that simplifies the design system in a sustainable and organized way.

Finally, design systems serve nor only as visual guidelines, but as a framework for workflows and communication. They improve a team's collaboration from a designer's completed sketches, prototypes, or components to the developers who will implement them into code (Yew et al., 2020).

2.3 Design Patterns

Pattern language, first introduced in the late 1970's by Christopher Alexander has influenced interaction design by supporting interdisciplinary collaboration among designers, developers and researchers. The three main advantages of pattern languages in interaction design are concretion, reusability across domains, and modifiability for generalization (Pan & Stolterman, 2013).

In software development, design patterns are object-oriented methodologies that offers reliable and reusable solutions to design challenges. Design patterns include elements like name, problem, solution and consequences, functioning as adaptable blueprints (Johri et al., 2022). Similarly, Preece et al. (2016) also describe patterns as a solution to a problem in a context, helping designers communicate and document experiences. Something in which Mirnig et al. (2016) mention in their paper: that a design pattern is a proven method for

designing solutions to reoccurring problems and that using of a pattern can improve the design process in terms of both time and effort.

For UI design, pattern libraries have become very useful as it provides solutions to common UI problems, reducing the need for manual research. These libraries sometimes include a pseudocode, which helps describe how a feature should be implemented (Chase, 2012). Kaneko et al. (2024) argue that design patterns act as “rule of thumb”, a way for designers to create effective user interfaces (UI) without having to “reinvent the wheel” for each project.

Christopher Alexander’s theory of “living structures” extends this idea. This idea promotes that any design that feels alive, harmonious and meaningful to people such as strong centers, contrast, gradients, echoes etc. (Stamey & Honeycutt, 2004). This perspective and approach, promotes the creation of technologies that support not only usability but also the user’s experience. It encourages designers to let the design evolve step by step and respond to the specific context and user needs as they emerge, aligning closely with user-centered HCI principles (Gorichanaz, 2023).

2.4 Design Guidelines

Design is both a product and a process, the product is a set of components and are shaped by resources and constraints, where the process involves various techniques and procedures to create the desired product, while the guidelines provide a foundation for effective design, though creativity remains essential (Johnson, 2020).

When designing a user-interface, rules and guidelines need to be followed: how people perceive, learn, reason, and act. A designer should have the perception in users mind when designing an interface. Perception is shaped by experiences, context, and goals. An example of this is if the positions of the buttons “Next” and “Back” swapped, users may press the wrong one automatically. Similarly, habituation happens when we gradually get used to repeated stimuli and stop reacting to it (Johnson, 2020).

To avoid ambiguity, designers should rely on user-testing as it verifies that the users interpret the design the same way. Designer can rely on standards or guide the user to interpret it as intended, ensuring consistent placement of elements like color, text, and fonts. Designers should also understand the goals; users often have a goal they want to achieve when they enter a system. As a user act as a main source of information about how things should be done, a designer’s assignment should be to communicate with them and give them an opportunity to try their versions of the developing design (Wood, 1997).

A user interface is usable when its functions are organized according to how users do their tasks, supporting task completion while enabling interactions that feel intuitive and natural. With the advancement in graphical user interfaces and system interface methods, alongside user-centered design, more usable applications have arisen. By focusing on usability, providing visible affordances and prevent user errors, designer can ensure that the design align with users’ natural way of interacting with tasks and can create interfaces that are both functional and intuitive, enhancing the overall experience (Wood, 1997).

2.5 Reusable and Sustainable Design

In the field of Human-Computer Interaction (HCI), sustainability has moved beyond energy efficiency and material to also include long-term usability, adaptability, and maintainability of systems. Within Sustainable HCI (SHCI), researchers argue that sustainable design is not achieved through isolated interventions but requires structured strategies and principles that support long-term impact (Knowles et al. 2016).

One central aspect of sustainability in design systems is reusability. In software engineering, design reuse is one of the core principles. It is through design patterns that reusable solutions are provided for common design problems. To improve reusability, König, Lang & Schaefer (2024) propose a concept that facilitates the use of these implementations by defining an abstract design pattern specification, that can be adapted to different contexts through specialized design decisions. They also point out the field's one-sided focus on understanding of sustainability, often reduced to energy metrics, and therefore call for clearer guidelines and definitions. Myers et al. (2000) points out that reusable UI components can be seen as a complementary approach that together enhance efficiency and promote long-term maintainability in user interface development. Tools help minimize the amount of coding required for developers when generating user interfaces. using a tool with a library of reusable components simplifies the process of creating and building user interfaces compared to coding them from scratch, while more recent work from Soares (2022) highlights that these libraries also facilitate collaboration between developers and designers. Most importantly, such tools will only be effective if it resonates with what the designers is experiencing: the way they think, solve problems and approach their work (Stolterman & Thandapani Selvan, 2008).

From a critical design perspective, Rachel Beth Egenhoefer (2024) argues that “sustainable design is not sustainable”, at least not in its current form. She critiques how designers are often romanticized as trendsetters, which contributes to the idea that “sustainable designers” are simply those who produce “sustainable goods”. Instead, she highlights that true sustainability involves creating designs that have durability, adaptability, a positive impact and a collaboration across multiple actors.

Finally, Fraser & Mancl (2024) highlight, that sustainability also emerges in the communication between people in a team, especially in the context of design and development work within large organizations. Sustainable software systems require more than technical competence, they depend on collaborative practices and adaptability across people systems and contexts.

2.6 AI & Automation in UX Tools

As the integration of artificial intelligence (AI) continues to evolve, it brings forward questions and challenges that remain unexplored in design contexts. Verganti, Venderaminelli and Lansiti (2020) argue, that in the context of AI, designers do not design solutions but design the problem solving. People tend to assume that AI simulates human behavior and thinking, but it operates differently. Most design decisions, for interface layout or information placement, need less imagination and more for specific problem-solving skills. In the context of AI, designers can avoid detailed development, and AI takes charge of creating specific solutions, through “problem solving loops”, that gather data, process it through algorithms,

and generates the optimal solution that is perfectly adapted for that single user. This, over time, leads to solutions being continuously improved and innovated.

AI-programs like ChatGPT-4o and Claude 3.5, suggest that AI also can recognize and reproduce visual patterns. Findings that resonate with Christopher Alexander findings on the importance of “living geometry”, patterns that reflect human emotional needs and cognition. AI despite not having human experience, can create and understand visual patterns that human recognizes as intelligence. With right feedback, AI can create patterns that create humans’ creativity and thinking, both crucial for effective user experience (Salingaros, 2025).

Future research could explore how AI supports design work and design systems. As intelligent technologies evolve, design methods for AI-driven contexts will also develop. Design systems may serve as a technical foundation, helping designers build cohesive, adaptable and ethically informed AI-driven experiences. Through machine learning, AI can even take typical design decisions such as interface layout, content creation and product positioning (Verganti, Vendraminelli, & Lansiti, 2020).

2.7 Perspectives on Design Systems

Previous research in UX design has highlighted both challenges and opportunities linked to design system. In Lamine & Cheng (2022) study, they found that design system are considered a bridge between designers and developers, that design systems should support both the visual UI editing for designers and the code base for developers. They highlight that the tools must resemble familiar workflows in the teams’ everyday lives to reduce learning barriers and that design systems should be adaptable to handle change due to evolution. Overall, they hope that their study can inspire others to investigate design systems.

Feng et al. (2023) argue that despite their research, some long-standing challenges remain. They investigated how UX practitioners experience the increased level of collaboration within tools such as Figma. The results showed that the tools create new opportunities for communication within teams, but despite this, other collaboration problems arise. The researchers therefore suggest that there should be an improvement that can provide support through text annotations directly in the design, so that the connection between design and code is improved. With this result, the researchers recommend how future tools and working methods can and should be improved to better and further support UX collaboration in practice.

3. Methodology

As we explore how a design system can be designed to ensure long-term usability and sustainability, we have chosen to use methods and frameworks suited to our research aim. This chapter presents our methodological approach, introducing with the theoretical frameworks guiding our study: *Research through Design* (RtD) and *Design Thinking*.

Thereafter, we present the structure of our data collection, which has been divided into three phases:

- **Phase A** focuses on gathering insights through interviews with experienced UX designers.

- **Phase B** involves the development of our design system, based on the insights from the interviews from Phase A, where we integrate both Research through Design (RtD) and Design Thinking as important components of our data collection, as they enable knowledge creation through the iterative and human centered design practices.
- **Phase C** centers on testing the developed design system and gathering feedback, as it is essential for understanding how well the design system functions in practice and for identifying potential improvements.

This structure reflects the actual process of our study, in which we progressed step by step from gathering insights to developing and evaluating our design system. The chapter also outlines our data analysis, how we aligned with the ethical research considerations and concludes with a reflection on our chosen methodology in relation to our research question and aim.

3.1 Research Through Design

Research through design (RtD) refers to design activities that actively contributes to knowledge. These design activities build upon common design methods and rely on designers skills, such as understanding problems and solving them and repeatedly creating and improving prototypes to address them (Stappers & Giaccardi, 2014).

By implementing this method in our research, we discovered new knowledge, as it relied on creating and testing design objects to understand problems and find possible solutions. We did this through weekly alignment meetings with our supervisor from Company X. These meetings also served as a primary source of data collection, as we gathered feedback, ideas, and answers to our questions, which allowed us to collaboratively create a design system that met their expectations (Olson and Kellogg, 2014).

RtD guided us by allowing us to discover new knowledge through the creation and testing of design objects. It focuses on improving the world, by creating novel artifacts that break or transform the current state of the world (Olson and Kellogg, 2014).

Our goal for the design system was for it to be functional, usable, and aligned with the rationalities and practical experiences that UX designers encounter in their daily work. By actively engaging in the design process and by creating and evaluating different versions of the design system, we ensured that our design system and component library continued to meet the needs of users and Company X. Not only did we create something functional, but we also generated new knowledge about how to organize, structure, and manage a design system and its components in a sustainable way.

Another important reason for using Research through Design as a framework was its emphasis on documentation. RtD allowed for structured and clear documentation of both the process and the outcomes, ensuring that others could adapt and build upon our system. While other researchers can reproduce the same process, there is no expectation that they will produce the same or even a similar final product. Because the design system will be used by developers and designers at Company X, we found it important to maintain transparent and straightforward documentation so that anyone could understand and utilize it (Olson and Kellogg, 2014). While our design process is described throughout this thesis, the detailed

documentation of our design system has been provided directly to Company X, in accordance with our collaboration agreement.

3.2 Design Thinking

Design thinking is an iterative process aimed at understanding users, critiquing assumptions, redefining problems, and creating innovative solutions that can be prototyped and tested. In other words, design thinking is a repetitive and a non-linear process, which involves continuous improvement and user-centered design principles. It focuses on understanding the users for whom we design for, enhances questions regarding the problem and is an ongoing process of testing new concepts and ideas. It contains of five phases: *1. Empathize, 2. Define, 3. Ideate, 4. Prototype and 5. Test* (Dam & Siang, 2024).

By applying design thinking in our research, we were able to carry out the assignment from Company X in a smooth and effective way. We believe that it simplified our design process due to its solution-based approach, focusing on investigating unknown prospects and uncovering previously unknown insights (Dam & Siang, 2024).

Since design thinking aims to improve services and products by analyzing and understanding how users interact with them, it encourages thinking outside the box. By exploring the conditions in which users operate, we gained deeper and valuable insights that helped us rethink challenges from multiple perspectives. In our case, we explored the conditions under which UX designers operate. As design thinking is a repetitive process, it required us to constantly question and develop knowledge throughout the process, which helped us redefine problems and identify alternative solutions that were not initially apparent to us (Dam & Siang, 2024).

3.3 Phase A – Interview

Our data collection must cover a broad range of topics: the tasks that users are currently performing and their associated goals, the context in which the tasks are carried out, and the logic behind the current situation. In the context of this study, this approach was essential to understanding the everyday realities of UX designers. By exploring their practical experiences and the rationales that guide their work, we were able to gather insights that contributed to how a design system can be developed to better align with their needs in real-world scenarios.

In this phase, data was generated through semi structured interviews. However, before initiating with our interviews, we had to develop an interview guide. An important step that ensures that our conversation would be relevant and aligned with the aim of our study. The following section describes the process behind the development of this guide.

3.3.1 Developing the Interview Guide

To understand how UX designers work with design systems in practice, we investigated their needs challenges and considerations by having interviews, more specific: semi-structured interviews. This was suitable within our research as this method allows for structure and flexibility. It enabled us to address pre-defined themes while still following up on the participants' interesting responses. These pre-defined themes were based on our research question and insights from earlier studies and literature on UX workflows. Even if we did not

have a pilot interview, the interview guide was refined and reviewed within the research team to ensure clarity and alignment with the purpose of the study.

These themes were structured around three areas: (1) Documentation and Guidelines: addressing how design systems are communicated and maintained; (2) Reusable and Sustainable Design: a focus on component structure, consistency and scalability; and (3) The role of the Design System: was exploring how the design system supports designers in practice and affects collaboration with developers. We included, in each theme, both open and specific questions, the open questions were more reflective, and the participants were allowed to think freely on their own experiences while the more specific questions were used to deeper identify key issues. The questions, to illustrate the nature of the interviews where “What do you see as the main advantages and disadvantages of working with reusable components?” and "How do you ensure that components work for users with different needs, such as those with motor or cognitive impairments?"

In total the interview guide included 15 main questions with support by follow-up questions to encourage the participants to elaborate. The questions were neutral to support reflectiveness and openness and aligned with the principles in human centered design (Kvale et al., 2014). The thematic and iterative process made us have an interview guide that gave us structure in the interviews that was also flexible to gather the participants perspectives and experiences.

3.3.2 Participant Selection

When it came to the selection of participants for our interviews, it was a combination of convenience sampling and purposive sampling. It was considered convenience sampling because it included available participants rather than specifically selected ones (Preece et al., 2016). But it was also purposive sampling because we deliberately selected participants based on specific criteria that made them relevant to the purpose of the study, i.e. they were all experienced UX designers. While one of our participants was connected to Company X, others were recruited externally and represented different organizations. Therefore, their insights reflected a broader range of professional contexts, rather than being limited to a single company’s perspective (Bullard, 2024).

There was a total of six individuals who participated in our interviews in Phase A. We chose to only conduct six interviews as we aimed for deep understanding rather than representativeness (Kvale, Brinkmann & Torhell, 2014). One of these six individuals was from Company X, as presented earlier in this section, because Company X only had one UX designer responsible for their design work in Figma. This made this person a key informant for understanding their design practices. By including this participant, we were able to gain in-depth insights from the company’s perspective, whereas the other participants, recruited externally, provided views from a broader range of professional contexts.

PARTICIPANTS	GENDER	AGE	BACKGROUND	TYPE OF EXPERIENCE	TIME INTERVIEW
<i>Participant 1 (P1)</i>	Male	49	bachelor's degree in Digital Media Production	3D Visualization Specialist and UX Consultant	1:06:47
<i>Participant 2 (P2)</i>	Female	24	MSc Interaction and Design	Worked with a design system during a summer internship.	0:27:22

<i>Participant 3 (P3)</i>	Male	29	MSc HCI & UX	Product/UX Designer & Design System Expert	0:26:56
<i>Participant 4 (P4)</i>	Female	23	MSc HCI & UX / bachelor's degree in Cognitive Science	Worked with a design system during a summer internship.	0:19:54
<i>Participant 5 (P5)</i>	Female	24	MSc Interaction and Design	UX/UI Designer	0:23:04
<i>Participant 6 (P6)</i>	Female	28	MSc Interaction and Design	UX/UI Designer	0:41:02

Table 1: Sample of participants in the interviews

3.3.3 Interview Procedure

Once we had selected participants for our study, we began by defining relevant subjects to cover and questions to be answered in our interviews, which can be viewed in Appendix A. To understand how UX designers work with design systems in practice, we investigated their needs challenges and considerations by having interviews, more specific: semi-structured interviews. Our questions were therefore both open and closed, the open-ended questions were more probing whereas the closed questions were asked when we knew the possible answer in advance. This approach allowed us to maintain a level of control through prepared questions, while also being open to exploring new topics that emerged during the conversation and delving deeper into them. The probing technique encouraged our participants to build on their thoughts, with follow-up questions such as “can you tell us more about that?” being asked to get the participants to think further (Preece et al., 2016).

Our interviews took place online through using Microsoft Teams, whereas the other ones took place face to face in a meeting room away from the interviewer’s usual workplace or home. In many cases, situations where the interview takes place in a different location can create an artificial context, i.e. that they feel separated from their normal activities. When this happens, it can be difficult for the participant to give detailed answers. To prevent this, the interview can be enriched by including prototypes, scenarios or descriptions of common tasks. In our case, we chose to interject some descriptions of different tasks. Given that the interviews were semi-structured, it therefore felt natural to interject an open question from time to time, whereupon the question could sometimes be somewhat more descriptive than others and lead to a discussion with the participant rather than an interview (Preece et al., 2016).

The advantage we saw in having most of our interviews online through Microsoft Teams was that some of our participants were unable to meet in person due to being situated in different areas. It was also beneficial for our participants, as they were able to choose whether to use video. For our study, the most important aspects were the collected data, the audio, the content of the interview, rather than seeing the participant or their appearance. This likely helped our participants to feel more at ease and relaxed, as they could be in their own environment without having to worry about their appearance or interacting in an unnatural setting with strangers (Preece et al., 2016).

3.3.4 Data Recording and Transcription

With the aim of improving the quality of the interviews, we recorded each interview. Having both recordings and transcriptions ensured that our analysis was based on what was said rather than memories and notes. This also enabled us to concentrate on the conversation and

unclear points, without the distraction of taking notes. We made sure to inform the participants of our intentions to record and secure their consent before, the consent of agreement was in line with our research ethics, which can be viewed in Appendix C. We understood the importance of our participants being comfortable with the recording process, therefore, we took measures to inform them that their anonymity would be maintained, and that they were aware of their right to withdraw at any time without any consequences. Additionally, we emphasized that the recordings would only be used for the purpose of our research and would be handled with the utmost confidentiality (Braun and Clarke, 2006).

A tool we used to transcribe our collected data was Microsoft Teams. Using that tool helped us optimize the process and it saved us time. For every interview, we started a meeting in Microsoft Teams. Regardless of whether the interview was conducted in person or online, we initiated a meeting to use the transcription function in the software. This way, we ensured consistent documentation of the interviews, which supported the analysis process.

3.3.5 Qualitative Analysis

Once the data collection sessions were completed, an initial processing of the data began. The analysis of the qualitative data is structured around a theoretical framework, as it led us to further insights that went beyond the results that emerged. Through the qualitative analysis, we have identified recurring patterns and themes and categorized the data, which can be viewed further in this section as an illustration. Through a categorization scheme, we were able to examine the nature of the data. It was after categorizing the data that a result emerged and we were able to answer the study's objectives (Preece et al., 2016).

Thematic analysis, as described by Braun and Clarke (2006), was applied for this qualitative method, as it provides a structured approach to interpret participants perspectives and addressing research questions. The method is used for identifying, analyzing and interpreting themes and patterns within data. This six-step process involves familiarizing with the data, coding, searching for themes, reviewing and refining them, and defining the final themes.

In practice, we transcribed the interviews into written form using Microsoft Team's tool. A step which is crucial for recognizing where meanings are created and in developing an understanding of the data. We then reviewed the material multiple times to recognize recurring patterns, initiating the code phase. Generating initial codes involves identifying and organizing a feature of the data that appears interesting for the analysis by coding specific segments of text. We applied this approach by working systematically through the whole transcription and gave equal attention to each part in the data and identified interesting aspects that may form the basis of repeated patterns. We coded by naming selections within each data set and organized it into meaningful sections (Braun & Clarke, 2006).

Through an iterative process, codes were organized into broader themes that captured important patterns within the data. Where some codes were combined, reorganized or excluded to ensure that themes accurately represent the data and its underlying meanings. We employed certain codes that were combined and restructured based on our reflections and discussions, when necessary, the codes that were no longer relevant to our research aim were excluded (Braun & Clarke, 2006).

Themes were refined, by revisiting the original data and comparing the identified themes with the interview transcripts. We discussed and refined the themes to ensure they were clearly defined and aligned with our research question and when necessary, we restructured the themes by breaking down broader themes into more specific subcategories. We took measures to understand and define our themes, ensuring that their titles clearly reflect their meaning, allowing the reader to immediately grasp what each theme is about (Braun & Clarke, 2006).

Finally, our findings were compiled into a structured report. The identified themes demonstrated their relevance to the research question and ensured that they were well-supported by the data. We systematically described each theme and outlined how it emerged from the analysis. To strengthen our findings, we included relevant interview excerpts as concrete examples. Additionally, we discussed the themes in relation to our research question and previous literature to provide a deeper understanding of their importance.

QUOTES	CODE	THEMES	CATEGORY
<i>"[...] It should be clear why a component exists and what its purpose is [...] But the most important thing is that it's built in a smart way so that it's easy to maintain"</i>	Good documentation / consistency / accessibility	Documentation for components & interactions	Documentation and Guidelines
<i>"It saves a tremendous amount of time because you don't have to reinvent the wheel every time [...] It's already been tested once, so I can be sure that what I pull from the design system or component library will meet the standards."</i>	Reusability / Saves time / Ensures quality	Efficiency & flexibility	Reusable and Sustainable Design

Table 2: Important quotes, codes and themes from the Thematic Analysis

3.4 Phase B – Development

With our chosen frameworks for the study, *Design Thinking* and *Research Through Design* (RtD), we established a structure that enabled an iterative process throughout the project. While Design Thinking is a human-centered and solutions-oriented approach (Dam & Siang, 2024), Research Through Design is a research method where knowledge is generated through the design process itself (Olson and Kellogg, 2014).

Both frameworks were influential in how we conducted our alignment meetings with our company supervisor at Company X. These meetings were structured to reflect the iterative nature of both frameworks, where we regularly presented our progress, received feedback and adjusted our design. We generated data through these sessions, not only from the outcomes of the design work but from the discussions, reflections and decisions made during the design process.

3.4.1 Participant Selection

For this type of data collection, it was evident that the most appropriate participant would be our supervisor from Company X. Consequently, this participant was selected through convenience sampling, as they were accessible within the context of our collaboration (Preece et al., 2016). In addition to serving as our supervisor, this individual was also responsible for the company's UX design and worked primarily in the design software Figma. Given the aim of our study, it was considered both reasonable and relevant within our research, to include this person as a participant. Since the RtD framework involves generating data through the

design process, it can also be argued that we, as researchers, became participants in this phase of data collection.

3.4.2 Data Procedure

As a result of our selected method RtD, we conducted weekly alignment meetings with our supervisor from Company X. We were able to get valuable feedback through these meetings and refine our design system iteratively. These refinements ensured that it met the practical needs and expectations of UX designers.

A preliminary version of a design system was presented to us at the beginning of our collaboration with Company X, that had been initiated within the company by our supervisor in Figma. We were encouraged to use this existing material as inspiration and to continue building upon it throughout our design process. This served as a starting point and reference, which helped ground our work in the company's actual design needs and context.

Through a signed consent form, our supervisor from Company X approved us from using these weekly meetings as a part of our data collection. The consent form, which can be viewed in Appendix C clearly stated our supervisors' rights and how we would handle and use the collected data in our research.

To make our data analysis process more efficient, the data was collected through notes. Instead of recording and later having to transcribe the meetings, we focused on taking notes on the most relevant subjects covered during our discussions.

In addition to our meetings with our supervisor, we as researchers also generated data through continuous internal collaboration. Together, the two of us created the design system, through regular discussions several times a week, where we reflected on insights, made design decisions, and evaluated our work iteratively. These internal dialogues played an incredibly important role in shaping our design system and are therefore considered part of our data collection.

3.4.3 Design Process and System Scope

By following the Design Thinking model, the development of our design system followed an iterative design process. These stages were empathizing, defining, ideating, prototyping, and testing, although when we did this in practice, these stages overlapped and was repeated as part of our RtD-driven workflow. With the results and insights from Phase A, during the empathizing stage, we could identify common challenges and needs among UX designers, which formed the foundation for defining our goals of the design system (Dam & Siang, 2024).

We created our design system using Company X's pre-existing file in Figma. Figma is a design tool who aim to make design available to everyone. One of their products is Figma Design, which allows people to create, share and test their designs for websites, mobile apps and other digital products. It is a tool used for and by designers and developers to help anyone involved in the project to contribute, provide feedback and make better decisions faster (Figma, 2024).

We explored possible directions for the design system during the ideation phase, by discussing potential components and searching online for inspiration. A process which was informed both by the empirical data from the interviews in Phase A, but also by ongoing feedback from our supervisor at Company X.

As we were given access to a preliminary version of a design system in Figma developed by our supervisor from Company X, the prototyping stage was carried out in Figma too. Their preliminary version was meant as inspiration. So, when prototyping, we divided our work equally and decided to first focus on the foundational elements of a design system, which was strongly recommended by both insights gathered in Phase A as well as from our alignment meetings with our supervisor from Company X. For everything to function properly later, we needed to spend a significant amount of time on establishing our tokens and variables. Tokens, as described in Figma's documentation (Figma, 2023), are a method for handling design properties and values across a design system and stores a piece of information such as color, font and sizing, to make them easier to refer to.

Our design system we created was not a complete or finalized system, but rather a scalable set of core components, intended to serve as a foundation that could be built upon within the company. It was agreed upon early in the process that the design system did not need to be fully finalized through an agreement with our supervisor from Company X. Instead, our design system was intended to be more of an initial structure that the company could later expand on internally.

The overall process of creating the design system meant that we had to focus on understanding our user, which is the primary goal when developing an interactive product, to understand your user. By categorizing the goals that the design system should cover, which in our case meant exploring how a design system can be developed to better align with what a UX designer experience in their daily work, we created a design system that aimed to meet both usability as well as a user experience goals. These goals included making the system easy to understand, efficient to use, aesthetically pleasing. This involved, among other things, making distinctions in the design of various components in terms of size and scale, and recognizing that designing for desktop and mobile required different strategies due to the variation in component dimensions (Preece et al., 2016).

3.5 Phase C - Test

To answer our research question of the study, we cannot solely rely on the interviews we conducted with the UX designers. We must, in addition, create a design system to understand how they can be developed to better align with the practical experiences that UX designers encounter in their daily work. It is through our chosen methods such as RtD and Design Thinking as well as alignment meetings with our supervisor and through regular internal discussions between us as researchers that we have developed a design system.

It is important to uphold usability and a positive user experience when developing a successful design system. Meaning, in interaction design, that the system is easy to understand, efficient to use and contribute to a meaningful experience for the user. By focusing on the user experience, the aim was to improve and expand in ways people work, this was in mind when we created the design system, that the system is easy to use and navigate in. To provide for this, an understanding of how people function, how they react in different situations and how they communicate is important when creating meaningful user experiences that are going to be used by several designers in a team (Preece et al., 2016).

We wanted the user tests to support iteration and refinement of the design system, not to be conducted to confirm a final product. By generating new and practical knowledge about how UX designers interact with and understand components and structures, we could understand how well the design system aligned with the real-world workflows.

3.5.1 Developing the Test Guide

To evaluate how well the design system worked in practice, we developed a test guide. The aim with the test was to simulate a work situation for UX designers and to test how well the design system worked in practice. The purpose was to understand how easy or difficult it was to use, if the components were interpreted as reusable and if the system matched actual work tasks. Rather than validating a final product, the goal was to We also wanted to get the participants' thoughts and interpretations of the design system after they tested it, so the participants were asked to answer a follow-up survey.

The test guide was therefore structured around three parts: (1) Background questions: asking questions about the participant's demographic information and relevant experience. (2) Scenario-based tasks: the participants acted as UX designers for a consulting company, testing their new client's early-stage design system. With this introduction, the participants were also given tasks focused on reusability and efficiency, with a focus on a design system to align with a UX designers practical experiences in their daily work. (3) Follow-up reflections: the participants were asked to evaluate their experiences through rating scales from 1 (very difficult) to 5 (very easy) and open-ended questions, such as "how time-efficient was it to make these changes?".

We developed the test guide continuously together with our supervisor at Company X, which was very helpful as the aim of this study is to explore how design systems can be developed in ways that support UX designers' everyday practices and long-term needs. This iterative collaboration ensured that the test reflected academic goals and relevance for the company.

3.5.2 Participant Selection

When it was time to select participants for our tests, the approach was, as in Phase A, a combination of convenience sampling and purposive sampling. The convenience aspect meant that we included participants who were available to us, while the purposive element ensured that the selected individuals were relevant to the aims of our study (Preece et al., 2016).

We conducted in total, four user tests, one in which was a pilot test used to evaluate and refine the test procedure itself. The remaining three participants took part in the refined user tests. All four participants had experience working as UX or UI designers, which meant that each participant had practical knowledge of design systems and regularly used tools such as Figma in their daily work. We affiliated one participant with Company X, while the other was recruited externally and represented different organizations, which therefore ensured a certain variation in background.

Based on our aim to gain in-depth insights rather than broad generalizability, we decided to only conduct three tests. In line with the principles of Research through Design (RtD), our goal was not to validate a finished solution through extensive testing, but to rather generate

situated knowledge by involving users in evaluating and reflecting on a design artifact that was still under development (Stolterman & Wiberg, 2010).

PARTICIPANTS	PHASE A & C	GENDER	AGE	BACKGROUND	TYPE OF EXPERIENCE	TIME TEST
<i>Pilot tester</i>	Only C	Female	24	MSc HCI & UX / bachelor's degree in Cognitive Science	UX Designer	0:32:00
<i>Test Participant 1 (TP1)</i>	Both A & C	Female	23	MSc HCI & UX / bachelor's degree in Cognitive Science	Worked with a design system during a summer internship.	0:25:00
<i>Test Participant 2 (TP2)</i>	Only C	Female	23	Bachelor's degree in digital media production	UX/UI and Graphic Designer	0:40:00
<i>Test Participant 3 (TP3)</i>	Both A & C	Female	28	MSc Interaction and Design	UX/UI Designer	0:35:00

Table 3: Sample of participants in the tests

3.5.3 Test Procedure

We chose to test our design system on participants through a form with instructions and questions, which can be viewed in Appendix B. As we aim for deep understanding rather than representativeness, we chose to only conduct four tests, where one of them were a pilot (Kvale, Brinkmann & Torhell, 2014). The pilot test helped us refine the instructions and rating scales before the remaining three tests were conducted.

All these four tests were structured through an online survey, because it felt most accessible for our participants with all the information collected in one place with instructions and questions. It also benefited us as researchers, by simplifying the data analysis process, by not having to make observations for instance. The online survey form followed this structure developed in the test guide but more refined after the pilot test: (1) demographics: questions about the participant's background and relevant experience. (2) Scenario: participants acted as consultants testing a startup's early-stage design system, where they were asked to evaluate it as UX designers, to see if the components and structure of the system aligned with real design tasks. (3) Tasks: participants completed predefined tasks presented as a checklist. For each presented task, they were instructed to mark a checkbox indicating that they succeeded with the task. If a task was not completed due to confusion or failure, they would leave the checkbox unchecked. (4) Evaluation: the participants were asked to evaluate their experiences through word-based scales from "very inefficient" to "very efficient" and answered open-ended questions, such as "how time-efficient was it to make these changes?".

Although there is no right or wrong way to arrange the numbers in a rating scale, some researchers suggest having 1 as the highest rating on the left and 5 as the lowest rating on the right. The reason being that the first place is the best in a race, and fifth place is the worst. Whereas other researchers prefer to arrange the scale the other way around. Preece et al. (2016) argue that a higher number intuitively indicates the best choice and the lowest number, the worst choice. In the pilot test it revealed that the numerical rating scales 1 (very difficult) to 5 (very easy) created confusion, so to reduce misinterpretation, we decided that it would be easier to just drop the numbers and instead use words: "very inefficient" to "very efficient".

3.5.4 Quantitative Analysis

Once the data collection in this phase was completed, we proceeded with the analysis of the collected material. This section outlines how the quantitative analysis was carried out, which was based on the user tests that concluded with a survey.

Although our user tests were limited to four participants (including one pilot), the structure of the test format allowed us to collect complementary quantitative data. The complementary quantitative data was: (1) Survey: used as a structure and support to systemize the participants' answers. (2) Quantitative elements: the checkboxes and rating scales were not treated as a generalizable result, but as indicators that complement the qualitative findings. (3) Selection: we purposely chose a limited selection of participants, as the aim was not to evaluate a finished product, but to gain practical and useful insights into how UX designers understood and used the design system, which was aligned with the principles of Research through Design (RtD) and Design Thinking.

We compiled the survey responses into a shared document for analysis. But before any analysis can take place, the data must be compiled into analyzable data sets. Typically, quantitative data is translated into rows and columns, where each row corresponds to an item, text respondent, or interviewee. By adding all of this to a spreadsheet, it becomes easier to filter the data set (Preece et al., 2016). However, in our case, as we chose to have a small sample size of participants, this process was straightforward and did not require any extensive categorization or advanced filtering. For this reason, we chose not to include any additional tables or appendix, since the dataset was limited and the patterns could be interpreted without any large-scale quantitative representation. Our shared document, acted as a practical tool to organize the results and ensured us that each participant's responses were systematically reviewed.

The quantitative elements in this phase were the checkboxes and rating scales, these elements and the results from them was not treated as a generalizable result, but as indicators that complement the qualitative findings. By combining this data with the insights from Phase A (interviews), we gained an understanding of how well the design system functioned in practice.

3.6 Ethical Considerations

Research is important and essential for individuals but also society. Therefore a legitimate requirement on how research should be conducted has been created by society and members of society, with a focus on fundamental issues and that it is of high quality. This requirement is called, the "research requirement", meaning that accessible knowledge is developed and thorough and that methods are improved (Vetenskapsrådet, 2002). Good research practice emphasizes the role in research ethics. It can be understood either as a set of ethical requirements or as the actual practice that arises when these standards are followed (Vetenskapsrådet, 2024).

As our research involves human participants, UX designers, we ensured so that research ethics are respected, as science need to be guided in in a way that foster trustworthiness, reproducibility and sustainability (ALLEA, n.d.). Researchers need to ensure respect for human dignity, minimize risks and balance them against expected knowledge. We ensure this

in our research by upholding and respecting ethical considerations that are often guided by the four key principles: beneficence (doing good), non-maleficence (avoiding harm), autonomy (respecting self-determination) and justice (Vetenskapsrådet, 2024).

Doing good in research means generating new knowledge that can be beneficial in various ways, referred to as the research interest. However, some of this knowledge cannot be collected without involving people, who may be exposed to risks that they need to be protected from, the protective interest (Vetenskapsrådet, 2024). Therefore, an essential principle of research ethics is that individuals need to be informed and give their consent before being involved in research. Getting consent is the most fundamental requirement, as it ensures and documents that participants were properly informed and made a voluntary decision to participate in the research. In our study, all participants were given a consent form to sign, as viewed in Appendix C, with details on our study's purpose, their rights, and the option to withdraw without any consequences. Therefore, our consent form met the following criteria: it must be *informed, voluntary, specific, explicit, and unambiguous, withdrawable, and documented*. Participants need to understand what their involvement imply (*informed*); they should not be forced or influenced to give their consent (*voluntary*); consent must be clearly defined so that participants can form a clear understanding of what they are consenting to (*specific*); there should be no hesitation whether participants have truly agreed to what they were told (*explicit and unambiguous*); participants can be able to withdraw consent at any time easily (*withdrawable*); and consent should be recorded properly and kept for as long as needed. The participants should also be offered to keep the information they need to understand what they have agreed to and how they can withdraw their consent (*documented*) (Vetenskapsrådet, 2024).

3.8 Methodological Reflections and Limitations

To validate is to examine, question and theorize (Kvale, Brinkmann & Torhell, 2014). Based on our research question “How can a design system be designed to ensure long-term usability and sustainability?”, we argue that by using a combination of methods: interviews, user testing and questionnaires and by applying the frameworks Research Through Design (RtD) and Design Thinking, we have been able to answer the question in a nuanced way in relation to the purpose of our study, which is “to explore how a design system can be developed to better align with the rationalities and practical experiences that UX designers encounter in their daily work”.

The interviews with experienced UX designers were a natural choice, given the research question. Through these interviews, we gained foundational insights into how designers work with and relate to a design system in practice. The interviews also provided us with valuable personal reflections on what role a design system has in a design process, including what they perceive as an advantage and disadvantage of using a design system, as well as the main challenges associated with maintaining one. However, it is of course possible to criticize our participants, the experienced UX designers that took part in our study. One could question their accounts and argue that what they shared may not be entirely truthful. On the other hand, in our study, we argue that the issue of validity is closely tied to the “what” of our research question. That is, we have investigated how UX designers experience and work with design

systems in practice. In other words, a subjective and experience-based perspective. Validity in this context is therefore not about proving an objective truth, but rather about gaining insights from the participants' perspectives in relation to the purpose of our study (Kvale, Brinkmann & Torhell, 2014).

On the other hand, what can be criticized is the fact that we did not include developers in this study. It would have been interesting to understand how the collaboration between designers and developers works by incorporating both perspectives on it. Even though we asked the UX designers about this collaboration, it would have been valuable to highlight the developers' side of it and see whether their answers were aligned or not. Additionally, developers might have responded differently to the other questions and given us alternative and potentially necessary insights.

When it comes to how we decided to test the design system, we chose to only obtain a total of four tests, one of which was a pilot test. In this part, we could have instead conducted continuous testing with our participants to increase reliability by enhancing the chances of identifying patterns in the user's (i.e., the UX designers) behavior but also the possibility of iterative improvement through feedback and evaluation, ultimately contributing to stronger validity.

However, it is with our chosen framework Research Through Design (RtD) that we have been able to improve the design system in an iterative way with the help of our supervisor from Company X. Through regular weekly meetings, we discussed and explored possible improvements to the system. In this way, we achieved a form of iterative work, even though we did not carry out several user tests with the participants. It is through engagement of the user (the supervisor from Company X) and the designer (us, who has created the design system) in the area, and through discussions about requirements, needs, expectations, and ambitions, that various insights have emerged regarding what was needed.

This process: the changes we made over time because of these alignment meetings is further detailed in our Result chapter, where we present the adjustments made throughout the project in collaboration with our supervisor from Company X.

This approach can certainly be criticized for not fully utilizing the users' feedback in several iterations. However, in relation to the study's aim "to explore how design systems can be developed in ways that support UX designers' everyday practices and long-term needs." we argue that the combined methodology: interviews, tests and close collaboration with a professional supervisor, has been sufficient to generate relevant data.

Finally, we are aware that our study is very empirical in nature, which may lead to limited generalizability. That is, that our study only focuses on a certain type of participants (experienced UX designers) and is conducted in collaboration with a particular tech consultant company based in Umeå (Company X). However, we argue that our conclusions are not limited to this specific case: rather, the results can be applied in other organizational contexts as well. The participants in our study are not exclusively from Company X, and therefore, their needs and ways of working are not unique to that organization. Instead, they come from diverse backgrounds and various organizations, which broadens the applicability of our findings.

4. Results

This chapter presents the findings of our study, structured in relation to the research question: “How can a Design System be developed to better align with UX designers’ practical workflows while ensuring long-term usability and maintainability?” and aim of the study: to explore how design systems can be developed in ways that support UX designers’ everyday practices and long-term needs. The results are organized in which insights were gathered, the design system was developed, and feedback was collected. To ensure anonymity, all participants are referred to as P1 to P6 and test participants as Test Participant 1 to 4, where Test Participant 1 refers to the participant that conducted the pilot test.

The result in this chapter is therefore structured in three phases which are reflecting the progression of the study.

- **Phase A** focuses on insights gathered from interviews with experienced UX designers.
- **Phase B** presents the developed design system based on insights from Phase A combined with frameworks: Research Through Design (RtD) and Design Thinking
- **Phase C** presents results on the developed design though user tests with participants.

This structure maintains a clear link between the insights that were gathered, the iterative design process and the evolution of the developed design system.

4.1 Phase A – Interview

These findings have been thematically organized into four main themes which are: *Documentation and Guidelines*, *Reusable and Sustainable Design*, *The Role of the Design System* and *Design Systems in a Future Context*. These themes provide an understanding of current challenges and considerations that are relevant when designing a usable design system.

4.1.1 Documentation and Guidelines

Having a well-structured design system is something that all participants agreed on. They believe that to achieve this, one should create coherent and accessible user experience. Something in which P6 emphasized, that components must be documented, consistent and grounded in a shared design system within the team. The components should be, not only visually and functionally aligned, but also accessible and adapted for various users’ situations. The participant also highlighted the importance of clearly defining different necessary states, to ensure that they function correctly across all types of backgrounds.

“I think it's absolutely well documented. That you have the various necessary states, for a component [...] it's important that the components are uniform and based on the same design system and that it's accessible [...] including basic stuff like that.”

- P3, interview, author’s translation

Similarly, P5 underlined, that documentation plays an essential role in encouraging a shared understanding of each components purpose and proper use. It is not enough for the component to just exist, they must be explained, contextualized and exemplified through real usage scenarios. For instance, the value of illustrating design patterns and providing clear guidance on when and how specific components should be used. P2 further emphasized that providing visual examples within the documentation can greatly aid in this process, helping bridge the gap between abstract guidelines and practical implementation.

“It should be clear why a component exists and what its purpose is. [...] But the most important thing is that it’s built in a smart way so that it’s easy to maintain.”

- P5, interview, author’s translation

Moreover, P5 described the benefits of reducing the amount of written documentation by instead working with clear structures. That the documentation should be accessible and understandable for everyone who interacts with the components, from designers to developers. To maintain an understanding in how patterns are described and ensuring technical smartness in how components are built is essential to achieve sustainability long-term.

“It’s important to follow the guidelines because it’s now also going to be legally required. But I think the most important thing is contrast.”

- P5, interview, author’s translation

This highlights that visual accessibility, particularly contrast between elements, must be considered and documented as part of the design process. P2 further mentioned that proper contrast ensures not only accessibility but also the readability and usability of interfaces across various devices. Another concern raised by P1 was the complexity of maintaining a component library as more designers contribute. She noted that even small, individual modifications can result in inconsistent over time if not managed properly.

“The more designers who use it, the more edge cases there will be [...] When everyone wants to make small changes all the time, the library may not be updated accordingly, which leads to inconsistency”

- P1, interview, author’s translation

To address this issue, P1 also advocated for assigning a product owner that are responsible and oversee the component library. This ensures that new components or changes are handled in a structured manner.

4.1.2 Reusable and Sustainable Design

According to the participants, the use of design principles and reusable components was crucial to ensuring both efficiency and consistency in design processes. The participants expressed a value in establishing a system for reusable components that is grounded in documented design principles. Such a system not only save time but also ensures the creation of digital solutions.

An important advantage of using reusable components was that it saves time. P5 emphasized that component libraries save a huge amount of time and ensure quality by allowing the team to avoid reinventing for each project.

"It saves a tremendous amount of time because you don't have to reinvent the wheel every time [...] It's already been tested once, so I can be sure that what I pull from the design system or component library will meet the standards."

- P5, interview, author's translation

Additionally, P5 highlighted the effectiveness of using tokens in Figma, particularly for ensuring a smooth design process. According to the participant, tokens offer significant advantages, especially when starting from scratch with a structured graphic profile. The flexibility of tokens allows for seamless updates, enabling designers to make changes in place that are automatically applied throughout the design system. Our findings are that the importance of working consistently with tokens and maintaining structure, even if building that structure takes time, it becomes worth it ultimately by enhancing efficiency and consistency of the design process. This view is shared by P2 who also pointed out how using pre-made component libraries optimizes prototype design and saves time, particularly in educational contexts where low-fidelity prototypes often are created. Moreover, P6 emphasized the technical importance of maintaining consistency through foundational design settings.

P6 emphasized the importance of establishing a structured foundation early in the design process by setting up a comprehensive font and color system that is integrated of all components. By organizing these elements into variables, the design becomes more adaptable for future changes.

By implementing reusable components, design teams can shift their focus forward the more creative aspects of the design process, while basic and repetitive needs are addressed through established systems. Our participants highlighted the value of reusable components in promoting efficiency and maintaining consistency. P1 emphasized that one of the major advantages is the increase in time efficiency, as essential needs are already covered using predefined components. Moreover, reusable components ensure consistency throughout the design, helping to create a cohesive and consistent design. This perspective was reinforced by P2, who reflected on the essential purpose of a component library, stating:

"I think it should be consistent, because that's kind of the point of a component library, that it follows a clear design. And I think it should be easy to adapt and that there might be documentation so that, well, you know... when you're about to make a change, you know what it should be able to do."

- P2, interview, author's translation

This statement underlines the importance of both maintaining consistency and providing flexibility through clear documentation to support future changes.

While reusable components were seen as beneficial, some of our participants also expressed concerns about potential disadvantages. P1 pointed out that a dependence on standardized

components could risk limiting creativity and variation in design solutions, that designers may become less inclined to experiment with new ideas. In contrast, P3 emphasized that creativity can still be preserved if the component library is designed to be flexible and follows for a certain degree of customization. According to this view, the component library should be seen as a tool that facilitates rather than restricts creativity, supporting variation within a structured and consistent framework. In addition to emphasizing the need for creative flexibility within a design system, the findings also highlight the critical importance of addressing practical aspects such as contrast, text visibility and accessibility during component design. Factors like ensuring sufficient text contrast, designing appropriately sized buttons for ease of interaction, and adapting the visual tone to the context of use, such as maintain a more formal aesthetic in applications, were identified as essential for aesthetic quality with functional accessibility. This was taken in consideration when designing the system, as shown in Appendix F.

Moreover, the significance of inclusive design practices, particularly with users with color vision deficiencies and ensuring usability across a diverse range of deviceses and user groups, these considerations can be integrating to not only support consistent and efficient design but also promote accessibility and inclusivity.

4.1.3 The Role of the Design System

Our study revealed that the Design System played a fundamental and highly influential part in the design process. All our participants were in clear agreement that the design system, holds a highly important and central role, not only within the design process but also in a broader organizational context, as it provides the structural foundation upon which the organization is built.

“In the design process, it kind of becomes a foundation in designing a product, or this page, or functionality, or whatever it is that you’re designing.”

- P6, interview, author’s translation

As the purpose of this study is to explore how a design system can be developed to better align with the practical experiences that UX designers encounter in their daily work, we asked all our participants about the role of a design system in the design process, and they all agreed on that it plays a significant and central role. The quote from P6 emphasizes that the role of the design system is foundational in the design process. By describing it as a “foundation” the respondent points out how essential and central the design system is, not only in the development of specific products or functionalities, but also in how it shapes the overall design framework. This means that a design system is more than just a collection of reusable components; it functions as a structural foundation for both the design process and the organizations efficiency. Which was something that we had in mind during the development and creation of our design system, as demonstrated in Appendix D.

The design system also plays an important role in facilitating the collaboration within a design team. A design system acts as a shared platform that helps ensure that team members are informed about how to work more consistently, which in turn leads to improved collaboration and reduced misunderstandings. By having such structure, the efficiency and

clarity of collaboration improves, which in turn leads to all team members from designers to developers making similar design choices.

“It becomes much easier, instead of us having to have a lot of dialogue with the developers [...] they can look in one place and understand that what is here is what we are going to produce.”

- P4, interview, author’s translation

Something in which participant 2 also highlights in their interview, the importance of consistency, stating that a design system provides a more unified approach, offering a reference point that helps reduce misunderstandings between designers and developers. As they put it, it creates:

“Something to work from and leads to fewer misunderstandings between designers and developers.”

- P2, interview, author’s translation

This means that the fundamental functionality associated with a design system, a component library, a collection of reusable buttons, is not its only primary focus and purpose. Rather, it lays the foundation for how an organization operates, collaborates and presents itself. Beyond its technical components, the component library serves as a shared resource that aligns team members, providing a common language and structure. As stated earlier, a design system enables a consistent design mindset when it comes to decisions and creation. Ensuring that all members of the team, are working with the same design principles.

4.1.4 Design Systems in a Future Context

One of our categories identified during the analytical process was design system in a future context. Our participants raised thoughts about what a design system could look like in the future and how it will be affected by new technologies and trends. The participants also raised thoughts about how to build a sustainable component library that will work long-term, which is what we presented earlier in this chapter under “5.2 Reusable and Sustainable Design”.

One thought that participant 2 raised was that in the future, tools like Figma, will have an AI feature that will simplify the process, the workflow around creating, maintaining and using a design system.

“I think it could be AI-adapted. But I hope it doesn't replace the role of a designer, it can perhaps simplify the process for a designer.”

- P2, interview, author’s translation

What participant 2 also expresses, is their concern of AI replacing the role of a UX designer. Something in which we noticed quite early in our data analysis, the common denominator among our respondents, AI. Many of our respondents, like participant 2, responded that AI could have a significant role in programs like Figma. Most likely a role that simply simplifies the structure and the process in a design system but also voicing their worry for it.

Participant 1 mentions in their interview, that an AI feature will possibly be available as a tool to categorize and manage the design system. But also, that there may be an innovative feature, that can produce several different variants of a color, for example.

“AI both as a UX thing to categorize [...] the design system, I think is good, then maybe an AI innovative thing to get for example variants of these blue ones [...] Then it generates variants of them.”

- P1, interview, author's translation

They continue and explain that they believe that we are now in a phase where technology is shifting. That something new needs to come, a new focus will appear, and they believe that tools that develops design systems will in some way, or another be a part of this shift.

“[...] I think we're on the way to some kind of technology shift. [...] a technical paradigm shift, everything will be changed. Then suddenly all content databases will be made towards a new focused area, which means that everything should be rewritten [...] AI will certainly be part of it. [...] Maybe the one that they will make will be too expensive to use because it has in and of itself. We and get on for everyone to do new things.”

- P1, interview, author's translation

When discussing trends, participant 5 mentioned that they hope we will go back to a more cheerful design. They expressed that today, design is so flat and lacks depth.

“I think everything right now is incredibly flat. I believe we might soon return to something more expressive, I hope so [...] But I'm not sure whether I truly believe that or if it's just wishful thinking. [...] There is so much you can do with AI it offers enormous potential. It can be used by developers and project managers, really in every part of the process. But I do feel that, as a designer, that function would be somewhat undervalued. Sure, what's being produced can be functional and even inspiring, but often it hasn't been directly usable in a meaningful way.”

- P5, interview, author's translation

They also empathize AI, but they don't see any risk with it, that it will somehow replace and take the role of a designer. AI may be utilized by developers or by project managers in their field, but specifically within the design area, they don't see any risk that there will be so useful that a designer will be replaced.

4.2 Phase B – Development of the Design System

The development of the design system is based on the insights that are gathered in Phase A as well as guided by frameworks such as Research through Design and Design Thinking. The design system was created iteratively by incorporating feedback from our supervisor and by discussions made internally to ensure that it aligns with both practical requirements and UX practices. Another part of this process was the scope of the design system, which was to focus on desktop platforms only. A decision that allowed us to prioritize refinement and long-term usability of components, ensuring that the design system could be thoroughly tested and iteratively improved within the available time and resources.

4.2.1 Iterative Design Process

Through our weekly meetings with our supervisor from Company X and internal discussions between us as researchers, we have been able to iteratively adjust and improve our design system. It is first and foremost through the results of our interviews that we began the creation.

Based on the results, presented earlier in this section, we argue that by creating a coherent and accessible user experience, one contributes to clear documentation that benefits everyone on the team. The process illustrated in Appendix E shows an example on how our design system evolved iteratively over time, emphasizing on generating knowledge through the creation and refinement of design artifacts (Stappers & Giaccardi, 2014).

Something that became evident during the project was that reusable and sustainable design is crucial to consider when creating a design system. Regarding sustainable design, many of our participants agreed that documentation and organization play a significant role, it is through ensuring these aspects that consistency is maintained and flexibility is enabled. By keeping a design system organized and structured, the design becomes more adaptable to future changes. Everything presented in this section was also therefore considered in the creation of our design system. Together with our supervisor from Company X, we were able to ensure this. This is visualized in Appendix F where we initially aimed to create a design system for both desktop and mobile platforms. But it was after a meeting with our supervisor that we concluded that to ensure long-term usability, it would be smart to solely focus on creating a design system intended for desktop, rather than have both desktop and mobile. Our supervisor considered it more time-consuming to include both; by focusing solely on creating a design system intended for desktop, we were able to spend more time refining and organizing the components to ensure long-term usability.

4.4 Phase C – User Testing of the Design System

Phase C presents the evaluation of our developed design system through user testing with UX designers. The purpose of this phase was not to validate a finalized product, but to test usability, effectiveness, and overall experience to identify strengths, areas for improvements, and opportunities for refinement.

EVALUATION ASPECT	FAIRLY POOR	FAIRLY POOR	NEUTRAL	FAIRLY GOOD	VERY GOOD
<i>Efficiency</i>	-	-	1	1 (2)	1
<i>Visual clarity & contrast</i>	-	-	1	2 (3)	-
<i>Component sizing</i>	-	-	(1)	2	1
<i>Flexibility in customizing colors and typography</i>	-	-	-	-	3 (4)
<i>Consistency of components and style guides</i>	-	-	(1)	1	2
<i>Overall usability</i>	-	-	-	1 (2)	2

Table 4: Summary of questionnaire responses from user testing

4.3.1 Test

To evaluate the usability and effectiveness of the developed design system, a pilot study was conducted. The aim was to gain initial insights into how the system's components and interaction supported users in understanding and completing specific tasks. The participant was asked to interact with the prototype and then asked to complete a questionnaire made to capture their experiences and perceptions.

The results from the pilot study provided valuable insights into both the strengths and area for improvements within the design system. The participant successfully completed most of the predefined tasks including customizing a component and testing it in light and dark modes. However, certain interactive states, like hover and focus, and more advanced adjustments were not completed, which indicating a need for clearer task descriptions and more specific instructions.

"I couldn't find the dark mode where I expected it"

- Test participant 1

These insights led to the decision to include more detailed instructions and a wider range of tasks for the participants to engage with. This allows for a broader evolution of the design system's functionality and user experience. The questionnaire responses reflected a mixed perception of the system's usability, while the participant appreciated features like the search function, they found aspects of the system less intuitive. Challenges were noted in locating certain components, such as the dark and light mode were not found were expected. Based on these findings, improvements were made for the main study, task instructions were adapted to more specific and to explicitly mention interaction states such as hover and focus. The test scenario was also expanded to include a broader and more set of tasks, aimed to encourage deeper exploration of the systems' consistency and usability.

Following the pilot test and the adjustments made to test the structure and instructions, the main study was conducted with a broader group of participants. The purpose of the main study was to evaluate the refined design system in a more structured and realistic context, assessing how well the updated version supported user tasks. With clearer task descriptions, the study aimed to capture insights into the usability and overall user experience of the system.

Three participants took part in the main study evaluating the developed design system. Their professional backgrounds were UX/UI designers contributing expertise knowledge relevant to the design systems intended context. Each participant was asked to perform a set of tasks in the design tool Figma, the tasks were adding and modifying buttons, testing component states and customizing typography and layout. These tasks were followed by a questionnaire capturing both quantitative responses and qualitative reflections.

As summarized in Table 4, the results reflected generally positive perceptions of usability, visual clarity, and efficiency. The table includes both the main study and the pilot study', with values in parentheses indicating responses from the pilot participant (e.g., "1 (2)" means one participant from the main study and two participants including the pilot selected that option). *Note: Original response options were presented in Swedish ("Mycket dåligt" to "Mycket bra").*

All participants successfully completed the tasks, including testing dark mode, adjusting component states and customizing text size and layout, this indicates that the system is functional and usable. Overall, the design system was perceived as easy to understand and use and the test participants described it as intuitive with components that were clear to search for and apply.

“The best part was that it was easy to search for and find components in the design system, and i was quick to make changes to them”

-Test participant 3

This perceived efficiency was further supported by another participant who highlighted how smooth it was to drag and drop components into the design without needing to start from scratch.

In terms of visual clarity and contrast, the feedback was generally positive. The test participants appreciated the clarity of the color choices and found the design visible accessible.

“Love the colors and contrast! I think it will make things much easier for those using the design system”

-Test participant 4

Even if ease of use were the overall response, there were several areas of improvements. One recurring issue was the difficulty in finding how to find the light and dark mode. Even though testing this feature was part of the task, the test participants needed to either search for the component again or experiment for a while until they figured it out.

“I had a bit of trouble figuring out where to switch from light to dark mode, but once you’ve found it once, it’s easy after that”

-Test participant 2

Another suggestion involved technical and visual consistency, for instance spacing between components was mentioned as an issue and the test participants noted that the sizes did not always align properly. When asked how well the design system supported UI states, such as hover and focus, participants responded positively.

The results indicate that the design system is perceived as user friendly, time saving and visually consistent. The participants valued the ease of locating and using components and appreciated the overall structure. At the same time, the findings point to opportunities to clarify certain functionalities and ensure alignment between the components. These insights from the questionnaire were valuable and important for further refinement of the system and indicated small adjustments that were made, such as adjusting the size and alignments in the buttons to better align with the other components and fixing the visual consistency in the design system. As the aim of this study is to explore how a design system can be developed to better align with UX designers work, the results, both the positive feedback and the areas of improvements, from the user tests were taken in high consideration and seen as valuable in the further refinement of the design system.

5. Discussion

Our study identified the following findings: *Documentation and Guidelines*, *Reusable and Sustainable Design*, *The Role of the Design System*, *Testing the Design System*, and *Design System in a Future Context* as well as insights gained from *the development and testing* of the design system itself. As the study aims to explore how design systems can be developed in ways that support UX designers' everyday practices and long-term needs, it is also essential to consider and analyze how these findings, in relation to our related studies, can deepen our understanding of current practices, reveal potential gaps, and contribute to the development of more user-centered design systems.

5.1 Design System in Relation to UX Practices and Prior Research

When it comes to deepening our understanding of current practices, potential gaps are revealed, which in turn may contribute to the development of more user-centered design systems. In our case, we observed through the results of our study, the importance that documentation and guidelines have in a design system. Consistency appeared as a crucial aspect in creating accessible user experiences, which aligns well with established research, emphasizing that design systems serve as a framework that support both efficiency and usability (Moore et al., 2020). Without a well-founded design system, inconsistencies arise quickly, which can potentially lead to poor user experiences and increased maintenance efforts, something in which the participants in our study also highlighted.

All our participant emphasized on the importance of documentation and consistency within a design system and understand how to navigate and proceed within it. This is most effectively achieved through proper documentation, as it supports not only direct communication but also sharing knowledge within teams, ensuring that knowledge, experience and collaborative skills can be passed on and developed within the organization over time (Fraser and Mancl, 2024).

On the other hand, it is important to remember that documentation should not replace communication and collaboration. Additionally, one insight from our participants was to balance the amount of documentation with clear structure and accessibility, they said that if you have too much text in a documentation, it can be overwhelming, while too little text leads to unanswered questions. By creating visual examples and descriptions helps make the guidelines more user friendly and easy to follow. Although a design system is considered a tool used to create consistent and high-quality software applications (Myllylä, Karvonen, & Koskinen, 2024), it must also be seen as a product, a system, that users use. In this case, UX designers and developers are the users. By understanding your user, you create a better product (Preece et al., 2016).

In our case, this means listening to our participants' feedback on what is considered too much or too little documentation. It is about finding a balance, combining both text and visuals, in a way that promotes good usability, while also achieving the user experience goals: that the design system should be easy to understand, efficient to use, and aesthetically pleasing (Preece et al., 2016).

With that in mind, having a well-documented design system with both text and visual content, the importance of having reusable components also surfaces. A reoccurring theme in our result was “reusable and sustainable design”, with sustainable not referring to the environmental impact it may have, although that is also an important aspect, in this context we rather mean how a design system can be created and maintained over time to be flexible, reusable, and adaptable to changing needs and work processes.

Several of our participants mentioned that reusable components not only simplify the work but also ensure a high quality in digital solutions. Something that is well consistent with established perspectives in HCI research, where design patterns and reusable components are seen as easy to manage and use over time (Knowles et al., 2016; Johri et al., 2022). Our participants also appreciated a feature in Figma: *Tokens*. These tokens act as a variable for color, fonts, spacing, and other design values, making it possible to quickly and consistently update the entire design system without having to change each component manually. This connects to Christopher Alexander’s idea of living structure in design, that systems change and grows over time but still is organized and evident (Knowles et al., 2016). The user tests supported this by showing that structured tokens helped designers change the components easy, which indicates that the design system is both adaptable and sustainable. At the same time, some participants argued that too much standardization could have an effect to limit creativity. While consistent design is important, using the same components too strictly can make it difficult to be innovative or adjust the design to new needs. This is argued by Johri et al., (2022) that design systems should be both structured and have flexibility. The user tests showed that designers prefer components that can be adjusted as needed, as long as the overall system stays coherent. A good design system should therefore, according to our findings support creativity rather than prevent it.

Sustainability in design systems is not only about technical artifacts but also about collaborative practices. Fraser & Mancl (2024) highlight, that sustainability emerges in the communication between people in a team, especially in the context of design and development work within large organizations. Sustainable software systems require more than technical competence, they depend on collaborative practices and adaptability across people systems and contexts. For UX designers, this means that a design system must not only provide reusable elements but also evolve in ways that reflect their needs and workflows. Otherwise, the system risks being optimized solely for developers, which undermines its long-term value for design practice.

Another important aspect that we considered essential to analyze, in relation to our related research was the emphasis on accessibility from the participants, particularly regarding visual contrast. For instance, when designing buttons, making sure these buttons are big enough to use as well as the design matches the context. These findings, were all seen as essential and they confirm the importance of intergrading usability principles into the design systems guidelines as well. Designing with usability for all users in mind, already from the beginning, can help prevent the need for changes later in the process (Gorchanax, 2023).

Our findings highlighted that accessibility should not be treated as an afterthought but rather as an integrated element that is documented with visual and functional components. This is consistent with the understanding that usability and aesthetics must coexist to create

interfaces that not only is aesthetically pleasant but work well for diverse user groups. The concern raised by participant that a design system is at its best when it is well documented and have accessible guidelines but also have organizational structures that support ongoing maintenance and evolution. By incorporating usability principles and providing clear documentation, design systems optimize design processes while delivering consistent and user-centered components.

As presented earlier in the result chapter, it is shown that a design system plays a crucial role in a design process. All participants in our study agreed on this. That a design system does not only function as a collection of reusable components but include the structure for how a design system is organized. This is confirmed in previous research, Myllylä et al. (2024), describing how a developed design system allows UX designers to work in a consistent and efficient way, especially in organizations where there are multiple members in a team. The design system serves not only as a practical tool for UX designers and developers but also as a framework with shared understanding that makes the work structured and consistent by having the same visual and interactive language. It becomes clear how important it is that a design system is not only functional but also adaptable for the long-term needs and methods that UX designers meet in their everyday work.

Stolterman & Selva (2008) highlighted that this type of tool, where the design system aligns with the designer's working logic and mindset, was clearly present in the participants experience. Several participants described how a design system creates security in the everyday work by establishing structured guidelines to support creative processes. In line with this, Moore et al. (2020) argue that the structure of the design system enables a design team to work more coordinated and faster because of the time of making decisions decrease.

When testing the design system, our test participants were given tasks to customize components, manage different states and adjust typography and layout. These tests, further confirms the design systems flexibility and usability, demonstrating that it effectively supports designers in making necessary adjustments while maintaining consistency across the interface. The usability in practice and the importance of understanding the design system is supported by Myers et al. (2000) that argues that UI tools must be powerful enough for advanced tasks but still simple enough to understand immediately, which was the case in our user test.

Even if the overall impression was positive, there was areas of improvements, such as difficulties in locating the dark and light mode. This result demonstrated the importance of clearer instructions and a more intuitive interface. Making it easier for users to find and change these settings. These results further reflect what Moore et al. (2020) found in their study, that even in robust design systems, discovering components can be a challenge.

Another aspect that we found was interesting to analyze, was emphasized in Soares (2022) article, the importance of having a clearly structured design system, demonstrating how the design system should be used, something that, in our case was not always self-evident. Design systems have overall an important role in structuring and optimizing the design process. Our study has shown that design systems is not only a good tool for reusing components, but also for the organizing an organization's design process.

As the study aims to understand how a design system can align with UX designers' experiences, it is also essential to consider and explore how these systems can evolve and be influenced by future technologies. The results indicate that the participants perceive AI as an important factor for the future development of design systems. Some of the participants expressed hopes for how AI can simplify the work by generating color variants or help adjusting components. According to Verganti, Vendraminelli and Lansiti's (2020) paper, they mentioned that in the context of AI, designers do not design the solution but the problem solving. They mean that people tend to assume that AI simulates human behavior and thinking, but AI works in a different way from designers.

Even if some of the participants were positive regarding AI, some were concerned that AI could replace the role of a designer and mentioned that AI should be a support rather than a replacement, something that is also evident in Salingaro (2024) work of Christopher Alexander's theory, that AI can by processing human data, replicate visual patterns that are perceived as meaningful to humans which means that AI could act as design tool rather than a replacement. When AI is given the opportunity to generate interface, predict behaviors and analyze user data, the design system would have to be more adaptive and could work as a framework where humans and AI collaborates. By having structured components and visual patterns in line with Christopher's Alexander's principles, such as symmetry, strong centers and contras, AI can create that are experienced as a human. The results from the design systems tests show that while users found the developed design system easy to use and intuitive, some challenges remained, especially in visuality which means that the test participants had a hard time finding certain modes, like dark and light mode. This indicates that human insights are important, even in systems where AI could play a larger role in the future, at the same time, as Verganti, Vendraminelli & Lansiti's (2020) AI could recognize the problem and give solutions to make the design system intuitive. AI is therefore perceived both as a potential risk and a helpful tool in the development of future design systems. While the participants supported AI, specifically the part where AI could make repetitive and detailed decisions easier, the importance of the human touch, creativity and ethical responsibility is at the same time essential. As AI becomes more common in design work, future design systems will need to find a balance between that, the automation and human input, a balance that require further study and practical development.

6. Conclusion

The aim of our study was to explore how design systems can be developed in ways that support UX designers' everyday practices and long-term needs. By collaborating with experienced UX designers, we were able to identify key aspects of their workflow, needs, and challenges which resulted in a development of a design system that was more relevant and user-centered for UX designers in practice.

Through our study, we have seen a clear connection between a well-structured documentation and the ability to create a sustainable design system. To ensure long-term usability and sustainability a Design System must have a consistent structure and thorough documentation. It is through the results from our interviews, user tests, and weekly alignment

meetings with our supervisor from Company X and our analysis that we have concluded that companies, needs to have a well-structured design system with clear documentation, and reusable component, as it is central to supporting UX designers in their daily work, to save time by avoiding repetitive tasks. A design system lays the foundation for how an organization operates, collaborates and presents itself. Beyond its technical components, a design system serves as a shared resource that aligns team members, providing them with a common language and structure.

But how can a design system then be developed to better align with UX designers practical workflows while ensuring long-term usability and maintainability? Well, by staying consistent while still being flexible, one has succeeded in creating a sustainable design system. Something in which resonates with Christopher Alexander's theory of a "living structure", meaning that designers are encouraged to develop digital systems that are not only functional, but intuitive, user-friendly and aligned with human needs and behaviors. In our context, design systems need to support usability as it should support user experience for UX designers in their daily work (Gorichanaz, 2023).

By incorporating theoretical design principles into applied practice, our study provides guidance for designing systems that are not only efficient but can be adaptive over time. The developed design system along with the findings from this study can serve as an inspiration for UX teams that are aiming to create or improve their own design system, with an emphasis on reusability, sustainably and consistency.

6.1 Future Research

While this study provides valuable insights into a design systems development, there are some areas that remain to be explored in future studies. First and foremost, the study could provide insights into how a design system evolve, adapt and remain sustainable when organization grow and new technologies shift the system. It would also be interesting how design systems perform in larger teams where multiple members contribute and use the system. An important direction for future research also involves how the integration of AI could be affected in UX designer's daily work. AI could support designers by generating design choices such as layout variation, managing components or suggesting interface pattern based on a user's behavior. Future research could also explore how to support UX designers everyday work in a way that support human creativity rather than replacing it, that would be an interesting contribution in relation to AI.

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Appendices

Appendix A

1. Background and Experience

- What is your background and role in UX design?
- Have you previously worked with a component library?
 - If yes, can you tell us more about that experience? What worked well, and what were the challenges?
- Which design tools and technologies have you used to work with design systems and component libraries?

2. Second part

- What role do you think a design system plays in the design process?
- What are the most important qualities of a good design system? Please elaborate.
- How does a design system influence collaboration between designers and developers?
- What are the biggest challenges when implementing and maintaining a design system? How do you handle changes without disrupting existing systems?

3. Third part

- What do you see as the biggest advantages and disadvantages of working with reusable components?
- How do you ensure that components work across different contexts and interfaces?

4. Fourth part

- What key factors do you consider when designing buttons?
- What common mistakes do you see in button design, and how can they be avoided?
- How can buttons be designed to ensure both great aesthetics and functionality?

5. Fifth part

- What accessibility aspects should always be considered when designing buttons and other UI components?
- How do you ensure that components work for users with different needs, such as those with motor or cognitive impairments (e.g., visual impairments, etc.)?
- Can you share examples of accessibility challenges you have encountered and how you addressed them?

6. Sixth part

- If you were to create a new component library from scratch, what would you prioritize first?
- How can a component library be built to remain effective over time and adapt to new needs without becoming cluttered or difficult to maintain?
- What trends or technologies do you think will influence the future of design systems?
- In your opinion, what makes a component library truly successful? Please elaborate.

Appendix B

Utvärdering av Designsystemet

Vi undersöker användarupplevelsen av ett nytt designsystem som tagits fram inom ramen för vår masteruppsats. Du kommer att få testa olika UI-komponenter och därefter besvara följande frågor baserat på din upplevelse av att arbeta med designsystemet som vi skapat. All data är anonym och används endast i forskningssyfte. Testet genomförs i Figma och är på engelska.

Syftet med detta designsystem är: Att utforska hur ett designsystem kan utvecklas för att bättre anpassa sig till de rationaliteter och praktiska erfarenheter som UX-designers möter i sitt dagliga arbete.

Bakgrundsfakta

1. Alder?

Ange ditt svar

2. Vad är din yrkesroll?

Ange ditt svar

Nästa

Utvärdering av Designsystemet

Test

Du är en UX-designer på ett konsultbolag som ska börja använda ett nytt designsystem av en startup kund. Du har fått i uppgift att **testa företagets designsystem** och dess **komponenter**. Kunden vill veta om designsystemet och dess komponenter är **återanvändningsbara** samt hur det kan *utvecklas* för att *bättre anpassa sig* till de *praktiska erfarenheter* som UX designers möter i sitt *dagliga* arbete.

3. Här kommer en del uppgifter du ska utföra

- ☐ Ta fram en meny från Assets
- ☐ Testa och se om den fungerar i Dark-mode.
- ☐ Därefter, placera ut två stycken knappar: Primary och Secondary
- ☐ Placera dessa bredvid varandra
- ☐ Gör primärknappen till en hemknapp genom att skriva "Home" och sätta en passande ikon till vänster om knappen.
- ☐ Ta därefter bort båda ikonerna i den sekundärknappen och skriv "More information"

4. Hur tidseffektivt var det att göra dessa ändringar?

Mycket ineffektivt	Ganska ineffektivt	Neutralt	Ganska tidseffektivt	Mycket tidseffektivt
☐	☐	☐	☐	☐

Appendix C

Consent and information about personal data processing



UMEÅ UNIVERSITET

Page 1 (2)

Consent of personal data processing and information to data subjects (GDPR)

In connection with the thesis project: Designing For the Future – Exploring Scalability and Human-Centered Principles in UX design at the department of Informatics, the following personal information about you will be collected and processed:

General information such as name, age, gender and relevant data linked to the study object for our thesis project.

Your consent is needed for the personal data types mentioned above to be processed prior to the student work. More information can be found below. Umeå University is responsible for the personal data being processed. Contact information is Umeå University, 901 87 Umeå, registrator@umu.se, 090-786 50 00. Umeå University has appointed a data protection officer who can be reached at pulo@umu.se or 090-786 50 00. You can also contact the person responsible for a thesis project via the course coordinator for the master/master thesis courses in Informatics, spring 2022, Rikard Harr, rikard.harr@umu.se, 090-786 92 82.

With the support of the consent, your personal data will be processed *no later than 15 October of the current year or until the time when the student work has been approved.*

Your personal information will only be handled by authorized students and staff at Umeå University. You have the right to withdraw your consent at any time by contacting the department via, studentsupport.informatik@umu.se, 090-786 65 23. Please note, however, that a revocation of your consent does not affect the legality of the processing before the consent has been revoked.

You also have the right to contact Umeå University to obtain information about what information is processed about you, or to request correction, transfer, deletion, or limitation of your personal information. You can also contact the university's data protection officer by email pulo@umu.se. For more information on how the university processes personal data, see: umu.se/gdpr

You also have the right to lodge a complaint with the supervisory authority, the Swedish Privacy Protection Authority, if you think that we process your personal data incorrectly.

Do you agree to the use of your personal data in the manner described above?

- Yes No

Date:

Name:

Information about Umeå University's processing of personal data in student work

If you participate, certain information about you will be processed: *name, gender, age and relevant data linked to the study object for our thesis project.* This information will be collected through interviews, which will be recorded and accompanied by note-taking. These recordings will then be transcribed and analyzed. The interviews will follow a semi-structured format. The information or part of the information will not be linked to you, everyone is considered personal data in accordance with the EU Data Protection Regulation 2016/679 (GDPR). The reason why such data needs to be processed is that it is essential for understanding and analyze the perspectives, experiences and insights of the participants in the study

Umeå University is responsible for personal data for this processing and the legal basis for the processing of personal data is your consent in accordance with the EU Data Protection Regulation, Article 6 (1) (a).

The personal data will be stored at the university in such a way that, in addition to the responsible student, only authorized staff at the university will be given access to it. The data will be processed so that unauthorized persons cannot access it.

The analysis of the results will be provided to ~~TUNX~~ Innovation, but no personal data from the participants will be shared – only the compiled anonymized analysis and findings.

Your personal data will be processed during the entire student work that will take place during the spring semester 2025, but no later than 15 October 2025.

Documentation for student work, which includes your personal information, is screened after the student's grades have been reported.

According to the EU Data Protection Regulation and national supplementary legislation, you have the right to:

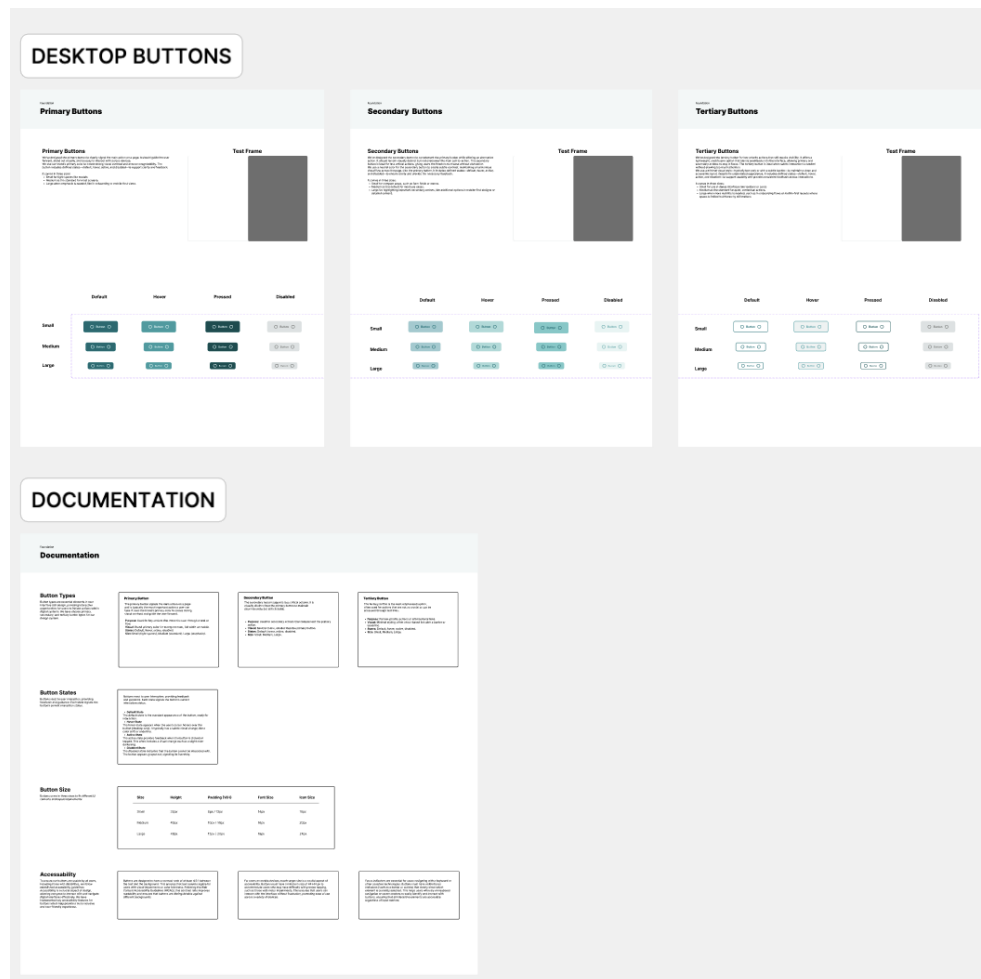
- Request access to your personal data (request so-called register extracts)
- Get your personal information corrected
- Have your personal data deleted
- Have the processing of your personal data restricted.

In certain circumstances, the Data Protection Regulation and supplementary national legislation allow for exceptions to these rights. The right of access to your data may, for example, be limited by confidentiality requirements, and the right to have data deleted may be limited by rules concerning archiving.

If you wish to invoke any of these rights, you must contact the Data Protection Officer at Umeå University (pulo@umu.se) and provide information on current student work.

If you are dissatisfied with how your personal data is processed, you have the right to complain to the Privacy Protection Authority. Information about this is available on the authority's website (imy.se).

Appendix D



Appendix E

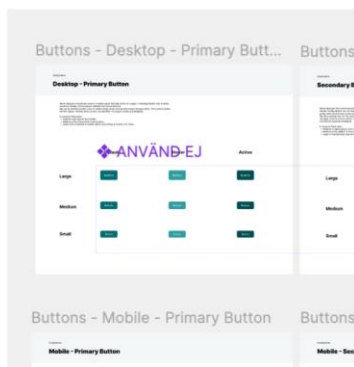
Before



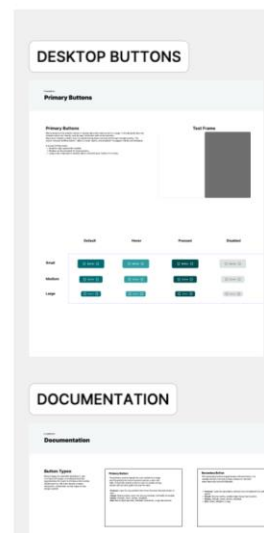
After



Before



After



Appendix F

Details

Thnx Innovation / Starting a new project

○ Button ○

Desktop Buttons

Includes 36 variants

Insert instance

Properties

Size

Large

State

Default

Type

Primary

Button Label

Button

Right Icon Type

holder

Left Icon Type

holder

Show Right Icon

Show Left Icon

Variable modes

Semantic

Light

Desktop

Headline

Ag

Headline 1 · 48/56

Ag

Headline 2 · 40/48

Ag

Headline 3 · 32/40

Title

Ag

Title 1 · 24/32

Ag

Title 2 · 20/28

Ag

Title 3 · 18/26

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subtitle · 14/20

Body

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Body 1 · 18/28

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Body 2 · 16/24

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Body 3 · 14/20

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